

September 2024

The HUMAN Digest 1.3

Lake Forest College's AI newsletter, courtesy of the HUMAN grant



Image created in Midjourney by David Sanchez-Burr

Greetings. In this issue you will find:

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HUMAN Digest 1.3

Welcome to the third issue of the HUMAN Digest. The pace of technological advancement in the field of AI is, at times, difficult to keep pace with and process. From generative models transforming how we interact with information, to AI-driven tools reshaping the classroom, the impact of these technologies on higher education is profound and accelerating. In this issue, we'll explore the ways AI is being integrated into college and other educational settings, not just as a subject of study but as a tool for enhancing teaching and learning. We hope to keep you up-to-date with articles relevant to higher education, as well as larger implications with respect to AI governance and downstream societal issues.

The HUMAN Digest welcomes feedback at kee@lakeforest.edu

Justin Kee, editor-in-chief

Davis Schneiderman, Principal Investigator, [HUMAN](#) grant

AI and the Humanities

Three case studies about the use of AI in the anthropology [classroom](#). Writing, diagnosing and “chatting” with a large database.

A European Union–backed project called [PERCEIVE](#) is using artificial intelligence to digitally restore the original hues of Edvard Munch’s work, *The Scream*, alongside other artworks, without physically altering the pieces.

The National Endowment for the Humanities recently awarded a number of [grants](#) to various colleges and universities to create research centers in the humanities for AI ethics, creativity and literacy.

Ray Schroeder, senior fellow at UPCEA, writes in his piece [“A Near-Future Vision of AI in Higher Ed”](#) on potential impacts of various technologies as they are rolled out on college campuses.

General news

The UN releases it’s [report](#) on governing AI for humanity. Key take-aways: the authors of the report are optimistic about the benefits to humanity of AI, there is a need for multiple stakeholders to be involved in crafting governance and regulations, and perhaps a need for a UN agency to establish international rules and norms for AI as we did for atomic energy.

Gavin Newsom, the governor of the State of California, recently signed a number of AI-safety related [bills](#). The various pieces of legislation cover digital “watermarking” of AI-generated content, sexually explicit AI-generated deepfakes, and to require social media companies to protect digital identities online.

Locally, [Microsoft](#) is investing \$3.3 billion into expanding a datacenter, creating an AI laboratory and providing training opportunities in Mount Pleasant, WI. As datacenters begin to scale up in size, [electrical power requirements](#) do as well.

OpenAI has released their latest models, [o1-preview and o1-mini](#), that the company claims can reason better about complex problems. This uses the novel approach of letting a model “think” for awhile before it begins generating text, which some commentators analogize to system 2 type thinking versus system 1 type thinking. These concepts were introduced by psychologist Daniel Kahneman in his book, *Thinking, Fast and Slow*.

Other perspectives

Author Ted Chiang [argues](#) that AI will never be able to make the art the way humans do, as it lacks the intentional decision making ability of humans artists.

Commentator Cody Fenwick [writes](#) about understanding digital minds and contemplating their moral status.

Sam Altman penned a [piece](#) on his vision for our future. Agree or disagree?

A useful term

Training In the context of machine learning, training refers to the process by which a computer learns to recognize patterns or make decisions from data. During training, a model is given a set of examples (called the training data), and it uses these examples to adjust its internal parameters so it can make better predictions or decisions in the future. This process is similar to how humans learn from experience: the more examples the model sees, the better it becomes at recognizing patterns. For example, if you are training a model to recognize cats in pictures, it will look at many labeled images of cats and non-cats and adjust its knowledge to improve its ability to correctly identify a cat in a new image.

People to know

Ilya [Sutskever](#) is a renowned computer scientist and one of the pioneers in the field of deep learning and artificial intelligence. He is best known as the co-founder and Chief Scientist of OpenAI, where he has led research into large language models and reinforcement learning. Sutskever's contributions to the development of neural networks are significant, particularly in co-authoring papers on machine learning techniques like long short-term memory (LSTM) and the development of the Transformer architecture, which has been pivotal for advances in natural language processing. Prior to OpenAI, he worked at Google Brain, where he played a central role in advancing Google's deep learning systems, including the development of models such as AlphaGo. With a Ph.D. in machine learning under Geoffrey Hinton at the University of Toronto, Sutskever has become a prominent

voice in AI ethics and was recently in charge of [superalignment](#) at OpenAI.

Ilya has recently left OpenAI to found a new company, Safe Superintelligence Inc., which has recently raised over a billion dollars in funding.

A visualization

Invented in 2017, the [transformer](#) architecture has led to the development of today's large-language models like ChatGPT. Here is a [visualization](#) of a part of the computational process inside where tokens “attend” to other tokens.

Just for fun

Take a look inside a Google [data center](#). Data centers are the key infrastructure behind cloud computing and now AI.

If you subscribe to the ChatGPT Plus account, you should have access to the real-time [voice integration](#) within the app. You can read about the capabilities [here](#).

HUMANS IN THE LOOP.

Have a tip or suggestion for an upcoming issue of the HUMAN Digest? Write Justin Kee an email at kee@lakeforest.edu.

